

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

Aim

To obtain exactly 2 gallons of water in the 4-gallon jug using a 4-gallon jug and a 3-gallon jug.

Problem Formulation

Initial State

- (0,0)
- Both Jugs are Empty

Goal State

- (2,0)
- The 4-gallon Jug contains exactly 2 gallons.

Operators

- Fill Either Jug.
- Empty either Jug.
- Pour Water from one jug to other until one is full or the other is empty.

STEP	4-GALLON JUG	3-GALLON JUG	ACTION
1	0	0	Initial State
2	0	3	Fill 3-gallon jug
3	3	0	Pour into 4-gallon jug
4	3	3	Fill 3-gallon jug again
5	4	2	Pour into 4-gallon until full
6	0	2	Empty 4-gallon jug
7	2	0	Pour remaining 2 gallons into 4-gallon jug

Goal Achieved:

The 4-gallon jug contains exactly 2 gallons.

2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.

- i. What are the percept's for this agent?
 - Camera images
 - sensor data
 - terrain
 - atmosphere
 - battery status
 - commands from Earth
- ii. Characterize the operating environment.
 - Partially observable
 - stochastic, sequential
 - dynamic
 - continuous
 - single-agent
 - unknown
- iii. What are the actions the agent can take?
 - Move forward, backward, and turn
 - Avoid obstacles
 - Capture images and videos
 - Collect soil and rock samples
 - Drill into rocks
 - Analyze samples using onboard scientific instruments
 - Measure atmospheric and environmental conditions
 - Store scientific data
 - Transmit data and images to Earth
 - Recharge batteries using solar panels (for solar-powered rovers)
 - Execute commands received from mission control
- iv. How can one evaluate the performance of the agent?
 - Successfully reaches exploration targets
 - Accuracy of scientific observations and sample analysis
 - Number of valuable scientific discoveries
 - Safe navigation with minimal collisions or damage
 - Efficient use of power and resources
 - Successful transmission of collected data to Earth
 - Ability to operate for a long mission duration
 - Reliability under harsh Martian conditions
- v. What sort of agent architecture do you think is most suitable for this agent? Why?

Reason:

A Mars Rover must make intelligent decisions in an uncertain and changing environment. A utility-based agent evaluates multiple possible actions and selects the one that maximizes mission objectives, such as scientific value, safety, and energy efficiency. Learning capabilities allow the

rover to improve navigation, adapt to new terrain, and optimize future decisions based on previous experiences.

Advantages:

- Adapts to unknown environments.
 - Makes optimal decisions by balancing scientific goals and resource constraints.
 - Learns from previous exploration data.
 - Increases mission efficiency and safety.
 - Handles uncertainty effectively.
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3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

Aim:

To formulate the 8-Queens problem and explore its search space using search strategies.

Problem Statement:

The objective is to place 8 queens on an 8×8 chessboard such that no two queens attack each other. Since a queen can move horizontally, vertically, and diagonally, no two queens should be in the same row, column, or diagonal.

Problem Formulation:

1. Initial State

An empty 8×8 chessboard with no queens placed.

2. State Space

- A state is any arrangement of queens on the chessboard.
- Queens are placed one column at a time.
- The search space contains all possible arrangements of 8 queens.

3. Actions

- Place a queen in the next column.
- Choose a row where the queen is not attacked by any previously placed queen.

4. Goal State

A solution is reached when:

- All 8 queens are placed.
- No two queens share the same:
 - Row
 - Column
 - Diagonal

Search Space Exploration

The search begins with an empty board and places one queen in each column. If a conflict occurs, the algorithm backtracks and tries another position.

Start (Empty Board) -> Place Queen in Column 1 -> Place Queen in Column 2 -> Place Queen in Column 3 -> ... -> Place Queen in Column 8 -> Goal State (All 8 queens placed safely)

Search Strategy Used

Backtracking (Depth-First Search)

- Place one queen in each column.
- Check whether the position is safe.
- If safe, continue to the next column.
- If no safe position exists, backtrack to the previous column and try another row.
- Continue until a valid solution is found.

Advantages:

- Reduces unnecessary search.
- Uses less memory than Breadth-First Search.
- Efficient for constraint satisfaction problems like the 8-Queens problem.

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4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

Aim:

To identify the suitable problem-solving agent used in an OLA Cab Booking application and write the pseudocode.

Type of Problem-Solving Agent:

The Goal-Based Agent is the most suitable agent for this problem.

Reason:

- The customer's goal is to travel from the source location to the destination.
- The agent evaluates available cab options (Mini, Micro, Sedan, Prime, Shared, etc.).
- It selects the cab that best satisfies the user's preferences, such as comfort, fare, and availability.
- Once a cab is selected, the agent books the ride and guides the customer to the destination.

Problem Components

Initial State

- Customer opens the OLA application.
- Current location and destination are entered.

Goal State

- Customer successfully reaches the destination.

Actions

- i. Detect current location.
- ii. Enter destination.
- iii. Display available cab types.
- iv. Select preferred cab.
- v. Confirm booking.
- vi. Assign driver.
- vii. Navigate to destination.
- viii. Complete the trip.

Pseudocode

BEGIN

Input Source Location

Input Destination

Display Available Cab Types

(Mini, Micro, Sedan, Prime, Shared, etc.)

Customer selects preferred cab

IF selected cab is available THEN

Assign nearest driver

Confirm booking

Navigate to destination

Complete trip

Display fare

ELSE

Display "Selected cab not available"

Suggest another cab type

ENDIF

END

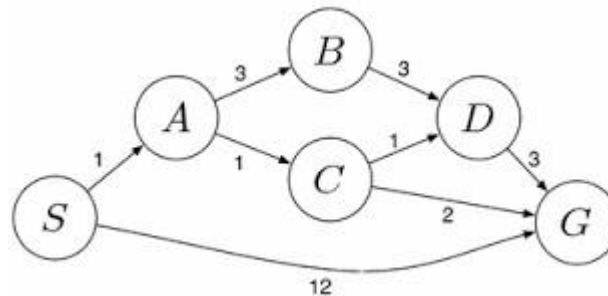
Conclusion:

The **Goal-Based Agent** is best suited for the OLA Cab Booking application because it focuses on achieving a specific goal—transporting the customer safely from the source to the destination while considering the customer's preferences and available cab options.

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5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform

Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Aim:

To determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

Given Graph:

The edge costs are:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

Step	Expanded Node	Path Cost	Frontier (Lower Cost First)
1	S	0	A-(1), G-(12)
2	A	1	C-(2), B-(4), G-(12)
3	C	2	D(3), G-(4), B-(4), G-(12)
4	D	3	G-(4), B-(4), G-(6), G-(12)
5	G	4	Goal Reached

Cost Calculation

Possible paths are:

1. $S \rightarrow G$
 - Cost = 12
2. $S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$
 - Cost = $1 + 3 + 3 + 3 = 10$
3. $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$
 - Cost = $1 + 1 + 1 + 3 = 6$
4. $S \rightarrow A \rightarrow C \rightarrow G$
 - Cost = $1 + 1 + 2 = 4$

Least-Cost Path = $S \rightarrow A \rightarrow C \rightarrow G \Rightarrow 4$

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem